

GIRINATH M

Dindigul, Tamil Nadu, India | +91 8122520060 | mgirinath05@gmail.com
[linkedin.com/in/mgirinath05](https://www.linkedin.com/in/mgirinath05) | github.com/mgirinath | <https://portfolio-s-taupe.vercel.app/>

CAREER OBJECTIVE

Electrical Engineering undergraduate bringing a unique blend of hardware-level systems thinking and software development skills in Java, SQL, and Python. Adaptable and quick to learn, seeking an entry-level software development role where embedded systems knowledge and programming ability intersect.

EDUCATIONAL QUALIFICATIONS

B.E., Electrical & Electronics Engineering, V.S.B Engineering College, Karur, Tamil Nadu | CGPA: 8.06/10 2023 – 2027
HSC, M.S.P. Solai Nadar Memorial Boys HSS, Dindigul | 74.83% 2021 – 2023

TECHNICAL SKILLS

Programming & Tools: Java, SQL, AutoCAD, MATLAB/Simulink, VS Code, GitHub, Embedded C++
Platforms: Arduino IDE, Blynk IoT Cloud, SAP Build (Joule Studio)

PROJECTS

Smart Ambient Humidifier

Technologies: ESP32 Microcontroller (Embedded C++), DHT22 Humidity Sensor

- Developed a humidity control system that monitors real-time ambient humidity levels and automatically regulates moisture using sensor feedback.

IoT-Automated Water Treatment Reactor

Technologies: C++ (Arduino), ESP8266 (NodeMCU), Blynk IoT Cloud, Serial Communication

- Built a cloud-connected water treatment simulation with automated treating, settling, and skimming phases; synced physical hardware with a real-time web dashboard and automated overcurrent safety interlocks.

PAPER PUBLICATION

IoT-Based Grass Cutting Robot, IEEE Xplore 2026

- Built a solar-powered, IoT-based autonomous grass-cutting robot; published at the International Conference on Innovative Computing (ICSES).

INTERNSHIPS

Industrial Intern, Tamil Nadu Newsprint and Papers Limited (TNPL) 2025 | Pugalur

- Gained hands-on exposure to industrial automation, renewable energy conversion, power distribution, and heavy electrical machinery.

Industrial Intern, Poppys Knit Wear Pvt. Ltd. 2025 | Kangeyam

- Explored industrial textile manufacturing processes and the integration of sustainable solar energy systems for industrial power.

WORKSHOPS

Potential Applications of AI & ML in Electrical Engineering, The Institution of Engineers (India), Tiruchirappalli 07/2025
Recent Advances in Hybrid Energy Storage Management for EV Applications, CSIR India 09/2024

CERTIFICATES

- Python Foundation Certification — Infosys Springboard
- Introduction to IoT and Digital Transformation — Cisco & NASSCOM
- Exploring SAP S/4HANA Sales Essentials — SAP Learning Hub
- Introducing Joule Studio in SAP Build — SAP Learning Hub
- Micro-Certification - Agentic AI Executive— Servicenow

ACHIEVEMENTS

- Secured Cash Prize in Paper Presentation, "SENSORICS 2024" — Kongu Engineering College, Perundurai, Erode (2024)
- Capgemini Brand Quest — Campus Ambassador, Capgemini (2024)
- NSS Member — Organised and led initiatives in college club activities

LANGUAGES KNOWN

Tamil, English

INTERESTS

Electric Vehicle (EV) Technology, Software Development, IoT Application Development, Embedded Systems